

machine learning

Supervised Lear. - PR-3.

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Q.1 What is logistic regression and why is it suitable for classification.

→ Logistic Regression is supervised machine learning algorithm used for classification problem. It predicts the probability that an input belongs to a particular class (e.g., Yes/No, Pass/Fail).

Why is it suitable for classification?

- It outputs probability between 0 and 1 using the sigmoid function.
- These probability can be converted into class labels (e.g., probability $> 0.5 \rightarrow$ class 1, otherwise class 0).
- It is simple, fast, and works well for binary classification tasks.

Example - Predicting whether an email is Spam or Not Spam.

Q.2 Explain classification performance metrics and why accuracy alone is insufficient.

→ Classification performance metrics are used to evaluate how well a classification model predict classes.

Common metrics:

- Accuracy - Percentage of correct predictions.
- Precision - How many predicted positive cases are actually positive.
- Recall - How many actual positive.
- F1-Score - Balance between Precision and Recall.

Why is its accuracy alone insufficient?

→ Accuracy can be misleading when the dataset is imbalanced. For example, if 95% of emails are not spam, a model that always predict "not spam" gets 95% accuracy but fails to detect spam. Therefore, Precision, Recall, and F1-Score provide a more complete evaluation.

Q.3. Define Type I Error and Type II Error in risk prediction content.

→ Type I Error (False Positive): Predicting that a risk exists when it actually does not.

Type II Error (False Negative): predicting that there is no risk when a risk actually exists.

Example (Disease Prediction):

- Type I Error: A healthy person is predicted to have the disease.
- Type II Error: A disease person is predicted to be healthy.

Q.4. Explain Precision, Recall, F1-Score, TPR, and FPR.

- Precision: Out of all predicted positive cases, how many are actually positive.

$$\text{Precision} = \frac{TP}{TP + FP}$$

- Recall (Sensitivity / TPR): Out of all actual positive cases, how many are correctly identified.

$$\text{Recall (TPR)} = \frac{TP}{TP + FN}$$

- F1-Score: Harmonic mean of Precision and Recall.

$$\text{F1-Score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}}$$

- TPR (True Positive Rate): same as Recall; measures correctly detected positives.

$$\text{TPR} = \frac{TP}{TP + FN}$$

- FPR (False Positive Rate): out of all actual negative cases, how many are incorrectly predicted as positive.

$$FPR = \frac{FP}{FP + TN}$$

where;

TP = True Positive, FP = False Positive, TN = True Negative, ~~FN = True False Neg Negative~~

Q.5. What is AUC-ROC and how does it help in evaluating classifier?

→ AUC-ROC stands for Area Under the Receiver Operating Characteristic curve.

- ROC curve - Plots True Positive Rate (TPR) against False Positive Rate (FPR) at different classification thresholds.
- AUC - Measures the overall ability of the classifier to distinguish between classes.

how it helps:

- Evaluates model performance across all thresholds.
- Higher AUC indicates better classification performance.
- AUC = 1.0 → Perfect classifier.
- AUC = 0.5 → Random guessing

so, AUC-ROC helps compare classifiers and assess how

well they separate positive and negative classes.

Q.6. Why does imbalanced data cause problem in classification models?

→ Imbalance data occurs when one class has significantly more samples than the other.

Why it causes problems:

- The model becomes biased towards the majority class.
- It may achieve high accuracy while performing poorly on the minority class.
- Important cases (e.g., fraud detection, disease prediction) may be missed.

Example: If 95% of transactions are normal, a model that always predict "normal" gets 95% accuracy but fails to detect fraud.

Therefore, metrics like precision, Recall, and F1-score are ~~pref~~ preferred for evaluating imbalanced datasets.